

Qatar-1b

R-2827

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_210801 · created 2026-09-05T23:16:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459427.739 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.054
Transit depth (Rp/Rs)^2	2.68 +/- 1.77 [%]
Orbital Inclination (inc)	83.06 +/- 2.56
Airmass coefficient 1 (a1)	0.865 +/- 0.026
Airmass coefficient 2 (a2)	0.12 +/- 0.16
Scatter in the residuals of the lightcurve fit is	3.01 %
Best Comparison Star	#3 - [289, 280]
Optimal Aperture	2.7
Optimal Annulus	4.89
Transit Duration (day)	0.058 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.054 vs 0.1463 (deviation 0.24σ)
Duration fit vs published	0.058 d vs 0.0692 d (deviation 0.37σ)
Mid-time O–C	1.3 min
Depth SNR	2.2
Residual scatter	3.01 %

Light curve

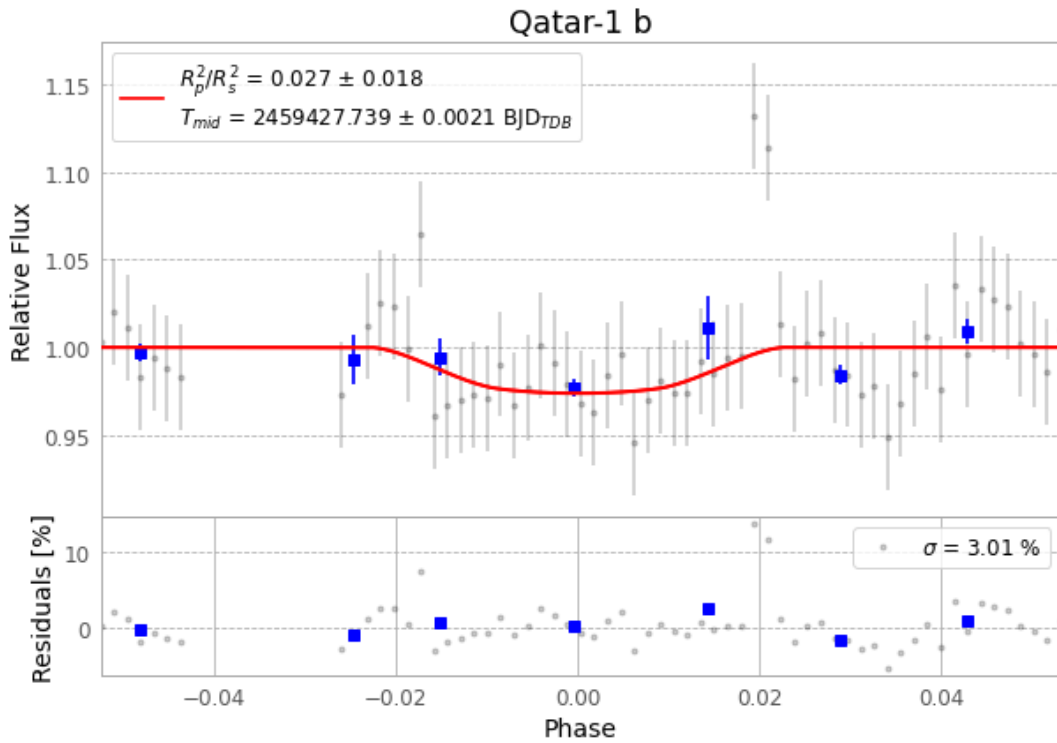


Figure 1 — FinalLightCurve_Qatar-1 b_2021-07-31.png (SHA-256 52f9c596193c2488...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [359, 259], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7f5c30659b541318...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

73 FITS frames (48 MB), Qatar-1210801035412.FITS → Qatar-1210801073012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-210801.log (719e9311d561...), FinalLightCurve_Qatar-1 b_2021-07-31.png (52f9c596193c...), FinalLightCurve_Qatar-1 b_2021-07-31.csv (f2144f5950dc...)

Compute resources

Wall time 565.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 23:16Z.